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REVIEW ARTICLE

Oral Chinese herbal medicine versus placebo for psoriasis vulgaris: A systematic review

Shefton Parker[a], Claire Shuiqing Zhang[a], Jason Jingjie Yu[a,b], Chuanjian Lu[a,b,c], Anthony Lin Zhang[a] and Charlie C. Xue[a,b]

[a]School of Health and Biomedical Sciences, China-Australia International Research Centre for Chinese Medicine, RMIT University, Melbourne, Australia; [b]Guangdong Provincial Academy of Chinese Medical Sciences and Guangdong Provincial Hospital of Chinese Medicine, Guangzhou, China; [c]Guangdong Provincial Key Laboratory of Clinical Research on Traditional Chinese Medicine Syndrome, Guangzhou, China

ABSTRACT

Objectives: Psoriasis is a chronic inflammatory skin disorder and the efficacy and safety of Chinese herbal medicine (CHM) treatments is unclear. This review evaluates oral CHM for psoriasis vulgaris clinical trial evidence.

Design: The Cochrane Library, PubMed, EMBASE, CINAHL, AMED, CBM, CNKI, CQVIP and Wanfang databases were searched from inception to June 2015. Randomised controlled trials (RCTs) of oral CHM compared to placebo data were included and analysed using Review Manager 5.2.

Results: Seven studies were included, no study utilised identical CHM intervention. Four studies data were subgrouped (decoction or capsule/pill) and pooled in meta-analysis to evaluate treatment effective rate for PASI60 or above (RR: 2.74 [0.92, 8.21] $I^2 = 65\%$). Another five studies were subgrouped and evaluated for PASI score change, (MD −7.00 [−10.74, −3.27] $I^2 = 98\%$). Only one study presented Dermatology Life Quality Index data, which favoured CHM (MD: −4.08 [−7.56, −0.60]). Two studies presented data on psoriasis-related inflammatory cell-signalling protein tumour necrosis factor-a (TNF-a) (MD: −4.92 [−5.31, −4.53]). No serious adverse events were reported.

Conclusion: While CHMs appear to be safe and may have benefit for psoriasis, variation between CHM interventions, outcome measures and the quality of included studies limit the conclusions of this review. Further rigorous RCTs utilising reliable, validated symptom and QoL outcome measures are recommended.

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Psoriasis; herb; random; control; trial; review

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Background

Psoriasis is a non-communicable chronic dermatological condition without current cure (1). Believed to be related to immune dysfunction, there is strong evidence of genetic involvement in its development; however, other potential triggers have also been proposed (2,3). The most common psoriasis phenotype is vulgaris (plaque type), presenting in around 80% of cases (4,5). Clinically, affected skin features raised and thickened areas, rough in appearance, erythematous with scaling on the surface (lesions) (6).

Severity is measured most commonly using the psoriasis area severity index (PASI) (7). Recently, emphasis has also been placed on improving the sufferer's quality of life (QoL) (8). Historically thought to be benign, emerging evidence associates psoriasis with the development of comorbidities, such as cardiovascular disease and diabetes (9). Treatments aim to reduce inflammation and decrease irregular skin cell proliferation (7). Recent discovery of biological drugs has seen improved antipsoriatic therapeutic effects; however, they pose risk of adverse effect, which are high in cost and potential for drug tolerance, and/or resistance can limit their use (10).

Chinese medicine has long history of treating psoriasis-related symptoms, with previous reviews identifying 174 different herbs have been used for psoriasis (11–13).

A review by Huang et al. 2012 was conducted on Chinese herbal medicine (CHM) for psoriasis; however, it consisted predominantly of literature review of classical CHM usage. While it did include a small section on clinical trial evidence, it was not comprehensive, searching only Chinese databases. The review did not appear to be conducted using a systematic approach and no risk of bias or meta-analyses were presented (14,15). Further outcomes were not those recommended in international treatment guidelines and adverse events were unreported (11).

A review by Zhang et al. (2014) evaluated oral CHM combined with conventional pharmaceutical treatment using systematic approach. Analysis of 14 randomised controlled trials (RCTs) ($n = 1279$) concluded CHM combined with conventional pharmaceutical (acitretin) was more effective than acitretin without addition of CHM ($n = 626$ RR 1.50[1.33, 1.70] $I^2 = 56\%$) for improving psoriasis (PASI and lesion score) (12).

Another comprehensive review of CHM RCT's of CHM bath therapy combined with phototherapy for psoriasis vulgaris included 13 RCTs ($n = 2168$). The results indicated combined therapy was superior to phototherapy without use of CHM (RR1.25 [1.18, 1.32] $I^2 = 23\%$) (13).

Evaluation of clinical evidence from RCTs is recognised as the most rigorous method to evaluate an intervention. Although previous reviews have been conducted for CHM efficacy, there is currently no published comprehensive systematic review comparing oral CHM to placebo for psoriasis vulgaris. In order to evaluate the effects of oral CHM for psoriasis vulgaris, this systematic review followed the Cochrane method and reported the results according to the PRISMA statement (14–16). Databases were searched for RCTs

CONTACT Prof. Charlie C. Xue, PhD charlie.xue@rmit.edu.au Executive Dean School of Health and Biomedical Sciences; Director, China-Australia International Research Centre for Chinese Medicine; RMIT University, PO Box 71 Bundoora, VIC. 3083, Australia

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of human subjects with psoriasis vulgaris receiving oral CHM compared with placebo. To determine efficacy, symptom severity scores were extracted from included studies. QoL, syndromes and adverse events data were also extracted to evaluate potential future use of oral CHM for psoriasis.

From laboratory study evidence, Chinese herbal medicine (CHM) contains compounds with potential antipsoriatic pharmaceutical like activity such as anti-inflammation and antiproliferation (17,18). In practice, CHMs are selected based on syndrome differentiation such as blood heat, blood dryness or blood stasis based on the synthesis of physical characteristics and symptoms (19). Ingredients of CHM for each included study were analysed for the frequency of use and phytochemical compounds of the most frequently used herbs were reviewed for antipsoriasis like activity.

Methods

Search strategy

The Cochrane Library, PubMed, EMBASE, CINAHL, AMED, CBM, CNKI, CQVIP and Wanfang databases were searched from their inception to June 2015. Search terms were grouped into three search blocks: 1. condition (psoriasis vulgaris), 2. CM intervention and 3. study design (RCTs) as synonyms of each. Example developed (for PubMed) for search terms were as follows: 1. Condition: Psoriasis or psoriases or psoriasiform or parapsoriasis or psoriatic psoriasis arthropathica or psoriasis, arthritic or skin diseases, papulosquamous or palmoplantar pustulosis or pustular psoriasis of palms and soles or pustulosis palmaris et plantaris or pustulosis of palms and soles or acrodermatitis continua or impetigo herpetiformis.

2. Intervention: Traditional Chinese medicine or Chinese traditional medicine or Chinese herbal drugs or Chinese drugs, plant or medicine, traditional or ethnopharmacology or ethnomedicine or ethnobotany or medicine, kampo or kampo or tcm or t.c.m. or medicine, ayurvedic or alternative medicine or complementary medicine or phytotherapy or herbology or plants, medicinal or plant preparations or plant extracts or plants, medicine or materia medica or single prescription or acupuncture or meridians or electroacupuncture or moxibustion or auriculotherapy or catgut embedding or herbs or Chinese medicine herb or herbal medicine.

3. Study type: Clinical trial or clinical study or biomedical research or controlled trial or controlled study or randomised control trial or randomised control study or multicentre study or meta-analysis or random allocation or double-blind or single-blind or comparative study or evaluation study or follow-up study or prospective study or research design or control group or placebo control or dummy control or blinding or clinical research or medical trial or *in vivo* study or case–control study or case study or intervention study or longitudinal study. A full list of all search terms for both English and Chinese databases is available upon request.

Study selection and data extraction

Randomised controlled trials data, where oral CHM was the intervention, were included. Condition was limited to psoriasis vulgaris (plaque) type, of any stage, severity or duration. Only trials published in English or Chinese were included with no limitations on age or gender. Available journal articles, conference proceedings and thesis were eligible for inclusion. Multiple arm studies were considered if one of the control arms was placebo. Studies investigating psoriasis type other than vulgaris or utilising CHM in a form other than orally were excluded. Any RCTs involving use of a conventional therapy along with CHM were also considered, providing the control group (placebo) received same conventional therapy and the therapy was recommended by international guidelines. Reviewers (Claire Shuiqing Zhang, Jason Jingjie Yu and Shefton Parker) extracted data from RCTs to Excel spread sheet. Data included author, year, country, trial design, number of participants, treatment duration, intervention, methodological quality, outcome measures, study results and adverse events. In situations of missing data, attempts were made to contact authors of publications to attain the information.

Risk of bias assessment

Risk of bias was assessed according to the Cochrane collaboration's tool by Claire Shuiqing Zhang, Jason Jingjie Yu and Shefton Parker for items of sequence generation, allocation concealment, blinding of participants and practitioner, blinding of outcome assessors, incomplete data, selective outcome reporting and other bias (if any).

Meta-analysis

Data were entered into RevMan 5.2 for analysis; clinically, heterogeneous studies were pooled together and within these meta-analyses, subgroups were formed of studies with similar characteristics. For dichotomous data, relative risk (RR) and 95% confidence interval (CI) were reported, while mean difference (MD) and 95% CI were reported for continuous data. Statistical heterogeneity was explored through sensitivity and subgroup analyses where the I^2 statistic was >50%. Where I^2 remained >50% a random effects model was employed and a sensitivity analysis conducted.

Results

Initial database search identified 16 020 records containing relevant search terms. After removal of duplicates and screening titles and abstracts, 1014 full-text articles were screened for inclusion, with data of seven identified RCTs from eight publications included (Figure 1). Two articles referred to data from the same study (20,21).

Description of the studies

Six studies were conducted in hospital outpatients, five of these, published in Chinese, were undertaken in China, two published in English where one was in Hong Kong (22) and the other did not state site location (23). The earliest study was published in 1999 (24) and most recent in 2012 (25). Two studies had a third arm (control other than placebo), but data were not included in this review (22,26). One study stratified participants, based on CM syndrome type, into one of three groups, yet all groups received same treatment (Table 1) (23).

Participants

Total number of included participants in the seven studies was 569, with 31 reported as either dropouts or lost to follow up, the remaining 538 were included in assessed data (Table 1).

Four studies were of participants with progressive stage psoriasis and remaining studies did not report stage (22,23,26). For severity criteria of inclusion, one study described participants as mild, moderate or severe (27), another included participants with body surface area (BSA) greater than 20% (22), while the remaining five did not state the severity for inclusion. Mean psoriasis

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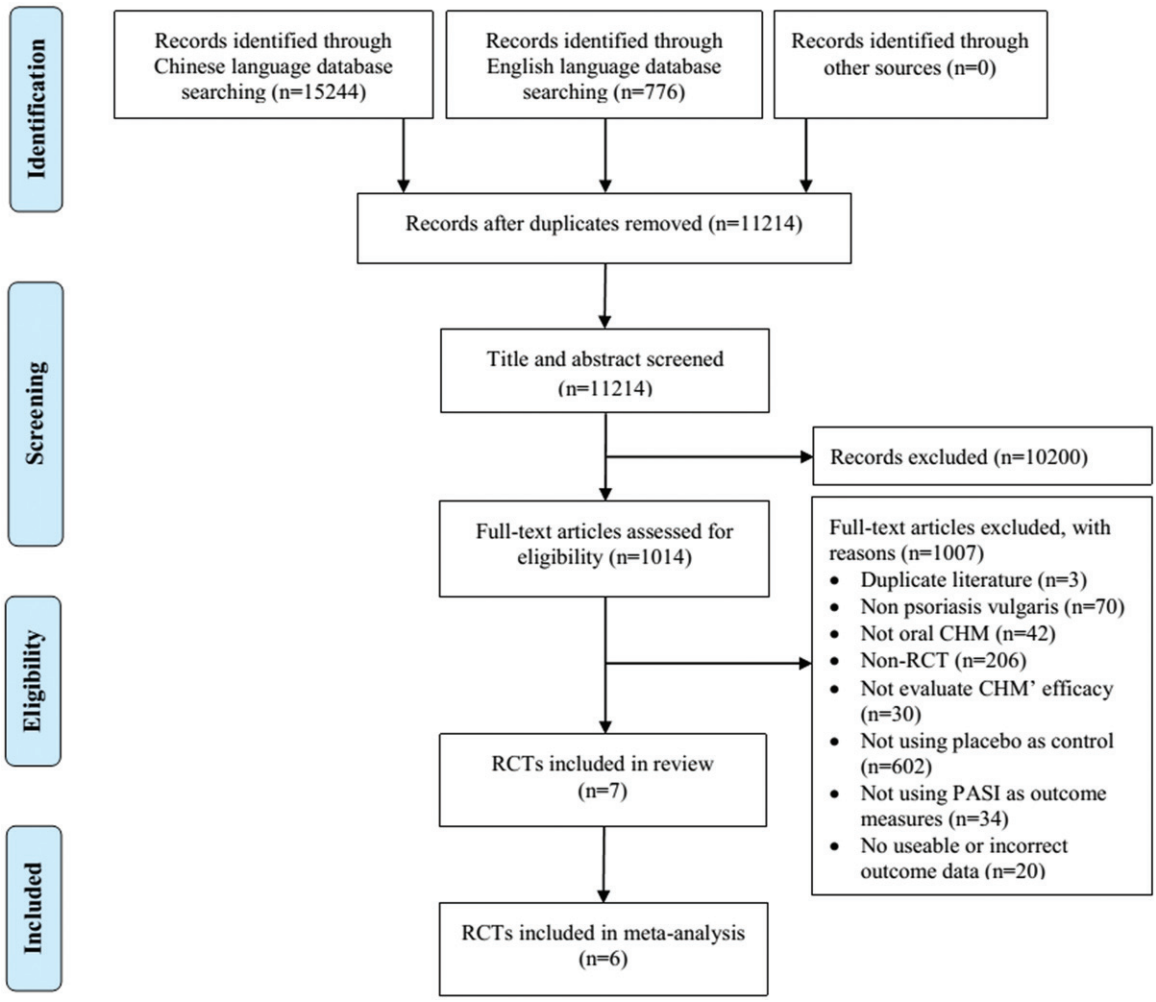

Figure 1. PRISMA flowchart of publication screening.

duration was not reported by Ho 2009, for remaining studies it ranged from 40 (21) to 106.8 (23) months in the intervention group and 39.6 (21) to 77.4 (23) months in control groups (Table 1).

Five studies provided the baseline PASI score for the intervention and control groups (21–25). Only Dai 2014 had a mean baseline PASI above 20 for the intervention groups at baseline (I1, I2 and I3 24.21 ± 2.49) the other four studies had mean PASI baseline scores below 20 (21,22,24,25). Two studies did not report the mean PASI score at baseline for the intervention group (26,27).

Interventions

Seven studies investigated multi-ingredient formulations, while one utilised single CHM, *Tripterygium wilfordii* (Lei gong teng) (24). This botanical has been indicated for use in rheumatoid arthritis (RA), where it reduces immune response inflammation and reduces T-cell proliferation, key aspects of psoriasis pathogenesis (28,29). Used in CM for conditions involving the immune system and inflammation, tripterygium wilfordii needs to be utilised with caution due to the toxicity of the leaves, flowers and skin of the roots (Table 2) (30).

Four of the studies utilised a decoction form intervention (23,25–27), the remaining three administered a capsule form (21,22,24). All studies utilised different CHM formulations, with a total of 47 different CHM utilised; however, some formulation ingredients were common between multiple trials. *Rehmannia* glutinosa was used in all multi-ingredient studies, although reportedly processed in two different forms. In raw form known as "*sheng di*" two studies utilised it (25,26) and in cooked with wine form, referred to as "*shu di*", it was in three studies (22,26,27). Dai 2014 did not report if processed or non-processed form was administered.

Salvia miltiorrhiza was an ingredient in all multi-ingredient CHM studies except Dai 2014. *Angelica sinensis* was utilised in four studies (23,25–27). The characteristics of each CHM are explored further in the discussion section. Four studies did not detail quantity of ingredients (21,23,24,27) and three did not detail complete ingredients of formulations (Table 2) (22,25,26).

Chinese medicine syndrome was utilised by four studies (23,25–27), remaining studies did not state if syndrome was used. Following syndrome differentiation, intervention type remained identical for participants in each study except Tan 2008, which administered one of four different formulations based on syndrome type (blood heat, blood dryness, blood deficiency or blood stasis).

All studies utilised placebo as a control arm, administered on its own in all studies except Dai 2014, which combined placebo and CHM arms with cod liver oil. Placebo was administered in capsule form for three studies (21,22,24), starch pill for one (27) and as decoction in two studies (23,25). One study did not provide

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Table 1. Characteristics of included RCTs (oral CHM vs. placebo).

First author; publication year; country; setting	Study design; blinding; number of arms	Treatment duration; follow-up duration	Stage; severity and duration of condition	Baseline PASI score	No. of participants randomised/assessed; dropouts	Age (mean (SD) or range); gender (M/F)	Intervention	Control 1	Control 2
Dai; 2014	RCT; NS; 4	8 w	I1:NS, NS, 9.62(5.33)y	26.27 ± 2.19	I1: 30/30; 0	I1: 38.7(11.8); 12/18	Yinxieling decoction and cod liver oil ointment	Placebo decoction (.5g caramel) and cod liver oil ointment	NA
			I2: NS, NS, 8.51(7.69)y	24.12 ± 1.08	I2: 30/30; 0	I2: 36.48(12.34); 11/19			
			I3: NS, NS, 8.59(5.4)y	22.23 ± 1.97	I3: 30/30; 0	I3: 36.3(12.7); 13/17			
			C: NS, NS, 6.45(8.15)y	24.83 ± 2.86	C: 30/30; 0	C: 37.39 (13.04); 14/16			
Ho; 2009; Hong Kong; hospital outpatient	RCT; S-B; 3	6 m; 0	I: NS; >20%BSA; NS	18.9 ± 8.2	I: 21/14; 7	I: 48.5; 14/7	Wen tong hua yu fang	Placebo capsule	Methotrexate and folic acid
			C1: NS; >20%BSA; NS	22.0 ± 11.3	C1: 20/17; 3	C1: 43.5; 18/2			
			C2: NS; >20%BSA; NS	20.4 ± 10.8	C2: 20/19; 1	C2: 38. 5; 18/2			
Su; 2010; China; hospital outpatient	RCT; NS; 2	12w; 0	I:progressive stage; mild; moderate and severe; 11.7 (3.5)y	NS	I: 35/35; 0	I: 48.8 (18.1); 28/7	Ke yin he ji	Placebo (Starch pill);	NA
			C:progressive stage; mild; moderate and severe; 11.4 (4.0)y	NS	C: 35/35; 0	C: 52.2 (16.1); 26/9			
Tan; 2008; China; hospital outpatient	RCT; S-B; 3	NS; 0	I: NS; NS; 6.3 (3.4)y	NS	I: 49/49; 0	I: 30.3 (7.23); 31/18	CHM (unnamed formula)	Placebo	Diyin tablet
			C1: NS; NS; 7.8 (4.7)y	NS	C1: 40/40; 0	C1: 31.6 (7.3); 27/13			
			C2: NS; NS; 7.3 (3.8)y	NS	C2: 47/47; 0	C2: 33.3 (6.4); 31/16			
Wang; 2003; China; hospital outpatient	RCT; D-B; 2	4w; 0	I:progressive stage; NS; 56.7 (40)m	9.07 ± 3.3	I: 27/25; 2	I: 26.7 (12.3); 16/9	Fu fang qing dai jiao nang	Placebo	NA
			C:progressive stage; NS; 61.2 (40)m	10.04 ± 4.47	C: 32/27; 5	C: 28.6 (14.2); 20/7			
Zhang; 1999; China; hospital outpatient	RCT; D-B; 2	4w; 0	I:progressive stage; NS; 56.7 (40.1)m	9.07 ± 4.23	I: 29/25; 4	I: 26.7 (12.3); 16/9	Lei gong teng jiao nang	Placebo capsule	NA
			C:progressive stage; NS; 61.2 (39.6)m	10.04 ± 4.47	C: 32/27; 5	C: 28.6 (14.3); 20/7			
Zhou; 2012; China; hospital outpatient	RCT; D-B; 2	8w; 0	I:progressive; NS; 4.0 (5.9)y	10.30 ± 2.65	I: 35/33; 2	I: 34.1 (9.2); 16/19	Liang xue huo xue fu fang	Placebo decoction	NA
			C:progressive; NS; 3.7 (5.8)y	9.08 ± 2.49	C: 27/25; 2 **T: 569/538/31**	C: 31.0 (7.9); 11/16			

B: blind; C: control; D: double; I: intervention; m: months; NA: not available; NS: not stated; RCT: randomised controlled trial; SD: standard deviation; ST: subtotal; w: weeks; y: years.

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sufficient detail to identify the placebo form, nor did it detail dosage or administration (26). Su 2010 utilised placebo in a different form (pill) to its CHM (decoction). This study utilised same administration frequency for both arms; however, dosage differed. Remaining studies utilised same dosage and administration procedures in both arms (Table 2).

Treatment duration and follow-up

Treatment duration of studies ranged from four weeks (21,24) to six months (22). Only one study did not state treatment duration (26). Treatment durations of four (21,24) or eight weeks (23,25) were most common. No studies presented follow-up data post treatment (Table 1).

Outcome measures

Total effective rate (TER) was the primary outcome measure used for all included studies except Dai 2014 study, which only utilised PASI score. Rates were all based on PASI score change (%) except Tan 2008, which based primary outcome on percentage of lesion reduction (95% and 50%). The endpoint for PASI score change varied for each study. Four studies indicated PASI90 endpoint as greatest degree of improvement (21,24,25,27), with mid-level improvement considered PASI70 (21,24) or PASI60 (27). Zhou 2012 required greatest PASI reduction outcome with 95% and Ho 2009 study had lowest PASI reduction outcome with 75% (Table 4).

Variation also existed for lowest level improvement, with two studies evaluating PASI50 (21,24), one PASI25 (27) and one PASI20 (25).

Of the studies four specifically reported PASI assessment data (21–23,25), with remaining only reporting effective rates. Two studies presented QoL data, only one utilising the most recognised QoL outcome for psoriasis, the dermatology life quality index (DLQI) (31) as well as a nondermatologically specific QoL instrument, the SF-36 (25). Ho 2009 utilised the psoriasis disability index (PDI), a measure of QoL specific to psoriasis but not commonly utilised (32).

Potential biological marker changes were reported in two studies (22,23), with no data on relapse rates provided by any studies (Table 4).

Dropouts

Four of the seven studies reported dropouts, with Ho 2009 study reporting highest ($n = 11$), seven of these from the intervention group. Zhang 1999 study reported nine dropouts with five from the control group. Wang 2003 study reporting seven dropouts most from the control group ($n = 5$) and Zhou 2012 study reported two dropouts from each group (Table 1).

Risk of bias assessment

Two studies were assessed as "low risk" for sequence generation (25,26) and the remaining assessed unclear. Allocation concealment was assessed as "low risk" for Wang 2003 and Zhou 2012 studies; all others were unclear. Risk of bias assessments for blinding of participants and blinding of personnel for Ho 2009 and Su 2010 studies were assessed as "high risk", Dai 2014 was unclear for personnel and remaining four low risk of both categories. For blinding of outcome assessors all were rated unclear. For incomplete outcome, of the seven studies, only three (23,26,27) were assessed as "low risk", considering none reported dropouts, final data were assumed complete. The other four studies were rated as "high risk", as they did not include dropout data. There was no evidence to suggest bias so selective outcome reporting was considered "low risk" for all studies. Without available published protocols for each study manuscript, data could not be completely evaluated for selective outcome reporting, as a result it could be argued this criterion should be assessed as unclear (Table 3).

Efficacy

For outcome measure evaluation, Chinese guidelines recommend minimum PASI60 for treatment to be considered successful. Internationally recognised clinical guidelines commonly consider treatment success as PASI75 and that below PASI50 as treatment failure (33). As reported in a similar designed systematic review, a PASI reduction of at least 60% was selected as the outcome criterion for meta-analysis and pooling of data (12).

Of those studies evaluating CHM compared to placebo four studies ($n = 215$) used an outcome of PASI60 or above (22,24,25,27). The four studies were subgrouped in either decoction form (25,27) or capsule/tablet form (22,24). Pooled together, the studies showed no significant effect for CHM intervention (RR: 2.74 [0.92, 8.21]), with considerable heterogeneity ($I^2 = 65$) (Figure 2). As separate subgroups, the studies using capsule/pill form still did not favour CHM (RR: 2.86 [0.01, 805.72]) and heterogeneity increased ($I^2 = 88$). However, the studies using decoction forms did favour CHM (RR: 2.65 [1.57, 4.49]) and the results showed no heterogeneity ($I^2 = 0$).

For PASI score change five studies (21–25) ($n = 141$) were separated into two subgroups, two using decoction form of CHM (23,25) and three using capsule/tablet form (21,22,24). When the subgroups were pooled together, they showed effect in favour of CHM, compared to placebo (MD: −7.00 [−10.74, −3.27]); however, heterogeneity of the studies was considerably high ($I^2 = 98$) (Figure 3). Pooled separately the capsule form studies were still in favour of CHM (MD: −7.41 [−8.34, −6.48]) and the heterogeneity significantly decreased ($I^2 = 35$). However, the two decoction studies failed to show significance as the confidence interval (CI) was very wide (MD: −8.10 [−17.89, 1.69]) and the heterogeneity of the studies was very high ($I^2 = 99$). The large difference in the baseline PASI severity score characteristics of participants between the Dai 2014 (24.21 ± 2.49) and Zhou 2012 10.30 ± 2.65 studies may explain the large CI and resulting heterogeneity.

Although not utilising PASI as an outcome, Tan 2008 study data ($n = 89$) also showed superiority of CHM compared to placebo, where CHM reduced lesions by at least 50% (RR:4.49 [2.40, 8.40]).

For QoL via the DLQI, only one study reported sufficient data (25), in which CHM showed improvement in health related QoL of subjects ($n = 58$) (MD: −4.08 [−7.56, −0.60]).

Known as a key inflammatory factor, tumour necrosis factor alpha (TNF-a) is often targeted by conventional drugs for psoriasis (34). Two studies (22,23), reported data on TNF-a change, showing CHM had effect on its reduction (MD: −4.92 [−5.31, −4.53]).

Safety

Adverse events data were presented in all but three studies (23,26,27). Gastrointestinal upset was most commonly reported by three studies (21,22,24). Diarrhoea or loose stools were next commonly reported, by three studies (21,24,25). Other reported adverse events included infection and abnormal liver function (number of events not reported) (22), and one event of menstrual disorder (24). The placebo control arm of the Ho 2009 study also reported adverse events of infection and increased liver enzymes, yet again did not report number of events. Regardless of similarity

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Table 2. Intervention/comparators of included RCTs (oral CHM vs. placebo).

First author, publication year	Syndrome differentiation	CM principle of treatment	Chinese herbal medicine formula and ingredients	Preparation type and dosage	Pharmacotherapy (specify)	Dosage and administration
Dai, 2014	Wind heat; blood stasis; blood dryness	Remove heat to cool blood, promote blood circulation remove blood stasis, and nourish yin and blood.	radix rehmanniae recen, angelica sinensis, radix paeoniae rubra, ligusticum wallichii, radices lithospermi, curcuma zedoary, chloranthus spicatus, rhizome smilacis glabrae, smoked plum, liquorice etc	Decoction 100 mL po, bid	Placebo decoction (0.5 g caramel)	100 mL po, bid
Ho, 2009	NS	NS	Wen tong hua yu fang: Ephedra sinica 6 g, Radix Aconiti Lateralis Praeparata 10 g, Semen brassicae 10 g, Cortex Cinnamomi 3 g, Rhizoma Zingiberis 3 g, Cornu Cervi Degelatinatum 15 g, Radix Rehmanniae Preparata 10 g, Rhizoma Smilacis Glabrae 60 g, Cortex Dictamn 30 g, Rhizoma Imperatae 30 g, dan shen 15 g, Caulis Spatholobi 30 g, Radix Lithospermi 30 g, Flos Sophorae Immaturus 30 g, Radix Glycyrrhizae 6 g, Indigo Naturalis 6 g	Capsule, NS	C1: Placebo capsule; C2: Methotrexate and folic acid	C1: Po, NS; C2: Methotrexate po, 2.5 mg then increase, no more than 30 mg per week; folic acid 5 mg per day
Su, 2010	Blood stasis	Nourish blood and clear dryness, invigorate blood and remove stasis	Ke yin he ji: Angelica Sinensis, Radix Rehmanniae Preparata, Radix Paeoniae Rubra, Semen Persicae, Radix Sophorae Flavescentis, Flos Carthami, Herba Taraxaci, Radix Salviae Miltiorrhizae, etc.	Decoction, 150 ml, po, bid	Placebo (Starch pill)	3.75 g/(kg.d), po, bid
Tan, 2008	4 types: Blood heat; blood dryness; blood deficiency; blood stasis	Blood heat type: clear heat and cool blood Blood dryness type: nourish Yin Blood deficiency type: tonify Blood Blood stasis type: remove stasis	Blood heat type: Flos Sophorae Immaturus 30 g, Radix Lithospermi 15 g, Radix Paeoniae Rubra 15 g, Rhizoma Imperatae 30 g, Radix Rehmanniae 30 g, Radix Salviae Miltiorrhizae 15 g, Radix Scutellaria baicalensis 15 g, Gypsum Fibrosum 30 g; Blood dryness type: Radix Rehmanniae Preparata 30 g, angelica sinensis, 30 g, Radix Astragali 30 g, Radix Polygoni Multiflori 30 g, Radix Salviae Miltiorrhizae 30 g, Semen Cannabis 15 g, Cortex Paeonia suffruticosa Andr 15 g, Radix Ophiopogonis 15 g, Radix asparagi 15 g, Fructus Tribuli 15 g, Radix Glycyrrhizae 10 g; Blood deficiency type: Rhizoma Atractylodis Macrocephalae 30 g, Radix Rehmanniae Preparata 30 g, angelica sinensis 30 g, Radix Paeoniae Alba 30 g, Scierotium Poriae Cocos 30 g, Radix Astragali 15 g, Caulis Spatholobi 15 g, Fructus Tribuli 10 g, Radix Glycyrrhizae 10 g; Blood stasis type: Radix Astragali 30 g, angelica sinensis 30 g, Radix Paeoniae Rubra 30 g, Rhizoma Chuanxiong 30 g, Semen Persicae 15 g, Cortex Dictamni 15 g, Pericarpium Citri Reticulatae 15 g, Flos Carthami 10 g, Radix Glycyrrhizae 10 g	Decoction, one pack per day	Placebo	NS
Wang, 2003	NS	NS	Fu fang qing dai jiao nang: Indigo Naturalis, Radix Angelicae Dahuricae, Radix Lithospermi, Radix Salviae Miltiorrhizae, etc.	Capsule, 12 po, qd	Placebo	Capsule, 12 po, qd
Zhang, 1999	NS	NS	Lei gong teng jiao nang: Tripterygium wilfordii	Capsule, 6 (18 g)/d, po	Placebo capsule, 6 (18 g)/d, po	6 capsules (18 g)/d, po
Zhou, 2012	Wind heat and blood dryness	Cool Blood and invigorate Blood	Liang xue huo xue fu fang: Folium Isatis tinctoria 15 g, Radix Rehmanniae 30 g, Radix Scutellaria baicalensis 12 g, Radix Lithospermi 9 g, Radix Salviae Miltiorrhizae 12 g, Radix Paeoniae Rubra 6 g, Cortex Paeonia suffruticosa Andr 9 g, angelica sinensis 12 g, Rhizoma Smilacis Glabrae 30 g, Cortex Dictamni 9 g, Herba Schizonepetae 6 g, Flos Lonicerae 20 g	Decoction, one pack a day	Placebo decoction	One pack a day

Table 3. Risk of bias assessment of included CTs (oral CHM vs. placebo).

First author, publication year	Sequence generation	Allocation concealment	Blinding of participants	Blinding of personnel	Blinding of outcome assessors	Incomplete outcome data	Selective outcome reporting
Dai, 2014	Unclear	Unclear	Low	Unclear	Unclear	Low	Low
Ho, 2009	Unclear	Unclear	High	High	Unclear	High	Low
Su, 2010	Unclear	Unclear	High	High	Unclear	Low	Low
Tan, 2008	Low	Low	Low	Low	Unclear	Low	Low
Wang, 2003	Unclear	Unclear	Low	Low	Unclear	High	Low
Zhang, 1999	Unclear	Unclear	Low	Low	Unclear	High	Low
Zhou, 2012	Low	Low	Low	Low	Unclear	High	Low

Table 4. Outcome measures of included RCTs (oral CHM vs. placebo).

		Psoriasis lesions			Quality of life				
First author, publication year	Definition of effectiveness	PASI score reduction	Lesion reduction	PASI score	DLQI	Other QoL assessment	Biological markers	Relapse rate	Adverse events
Dai, 2014	PASI change	–	–	Yes	–	–	Yes	–	–
Ho, 2009	PASI 75, 50	Yes	–	Yes	–	Psoriasis disability index	–	–	yes
Su, 2010	PASI 90, 60, 25	Yes	–	–	–	–	Yes	–	yes
Tan, 2008	Lesion reduction (%) 95, 50	–	Yes	–	–	–	–	–	–
Wang, 2003	PASI 90, 70, 50	Yes	–	Yes	–	–	–	–	yes
Zhang, 1999	PASI 90, 70, 50	Yes	–	Yes	–	–	–	–	yes
Zhou, 2012	PASI 95, 60, 20	Yes	–	Yes	Yes	SF-36	–	–	yes

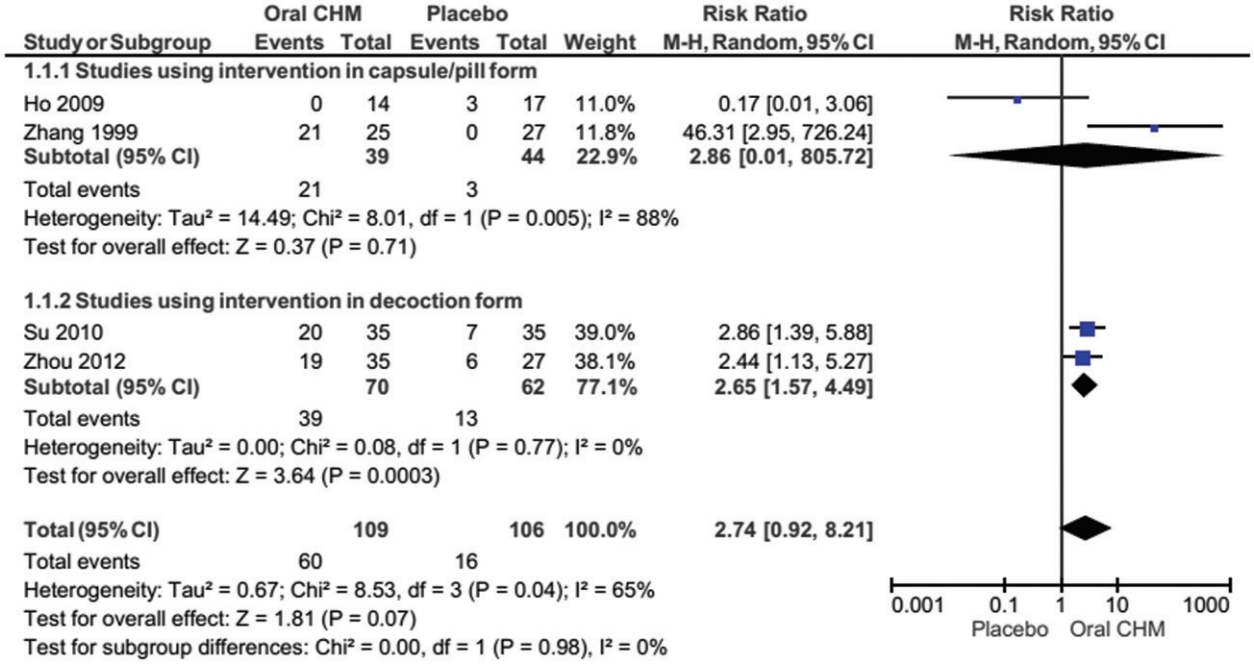

Figure 2. Meta-Analysis forest plot (PASI 60 and above).

of adverse events between Ho 2009 control and intervention arms they concluded tolerability of CHM is moderate.

Discussion

Efficacy of oral CM versus placebo

Although review found CHM significantly effects PASI change, CHM generally did not significantly reduce PASI score by at least 60% compared to placebo. Although the meta-analysis indicated CHM in decoction form is effective to reduce PASI by 60%, it is noted only four studies were eligible for PASI60 assessment involving a total of 215 participants, and the sample size was not sufficient to detect general significance of CHM.

Interestingly, the reverse was true for PASI score improvement with the pooled studies favouring CHM and subgroups, indicating CHM capsule/pill form to be superior to decoction form.

Each study utilised different herbs, future research should compare effects between the different forms (i.e. capsules vs. decoction) of same herbal ingredients to investigate their impact on outcomes. Only one study (25) utilised commonly recommended QoL outcome the DLQI, and found CHM had greater effect. To evaluate oral CHM for QoL, more data are needed. Future studies should include a reliable and validated QoL instrument for psoriasis.

Although results suggest oral CHM may improve psoriasis, it is unclear if there is a specific formulation providing greatest benefit. As each study's CHM formulation was unique, it is also difficult to

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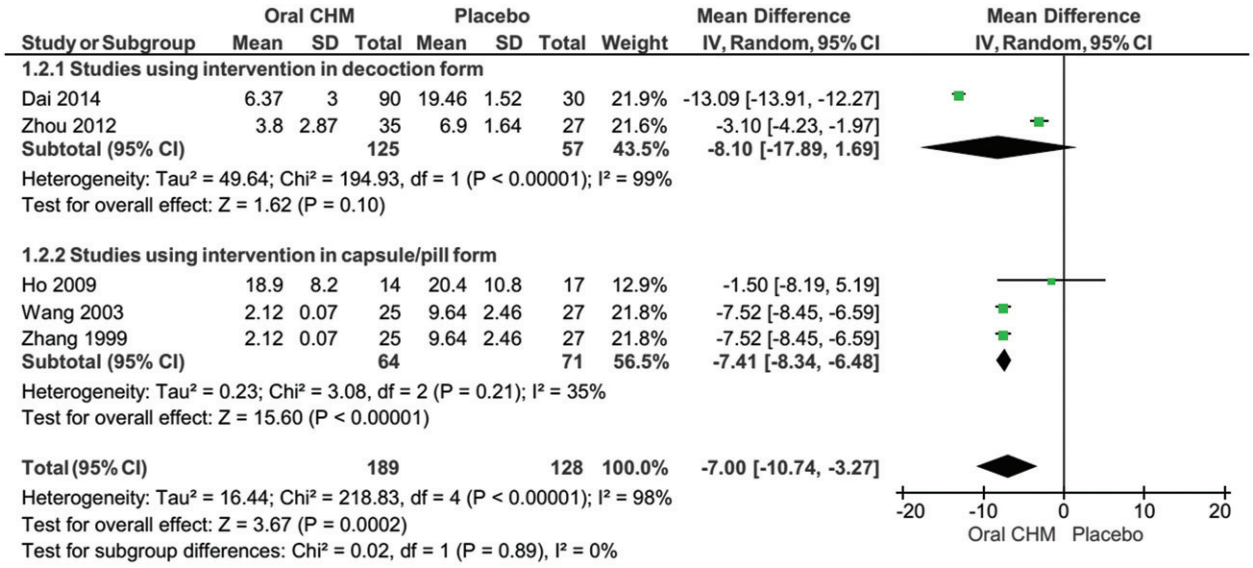

Figure 3. Meta-analysis forest plot (PASI score).

distinguish which ingredients give the largest therapeutic response. To better compare effect, future studies might consider additional arms, consisting of different CHM formulations. More pre-clinical evidence such as *in vitro* and *in vivo* research may provide data for researchers to construct an optimised psoriasis-specific CHM formulation that could be further explored in clinical trial (35).

With previous clinical and pre-clinical evidence suggesting its potential in immune disease, Zhang (1999) investigated single oral CHM *tripterygium wilfordii* for psoriasis (36–38). Interestingly, despite it having the greatest effect achieving PASI60, no other reviewed studies included it in their formulations.

Safety

All studies reported relatively mild or at most moderate side effects for oral CHM. Interestingly, from Zhang (1999), although considered toxic, *tripterygium wilfordii* reported only mild gastrointestinal disturbance, relieved by reduction in its dosage. It is highlighted that the preparation and processing of *tripterygium wilfordii* is important for reducing its toxicity (39). In addition to reduce toxic effects on the gastrointestinal tract, researchers and clinicians should consider combining *glycyrrhizae radix*, which is often added to reduce toxic effects of other ingredients (40).

Ho 2009 utilised two CHM well known for their toxicity, *ephedra sinica* and *radix aconiti lateralis praeparata*, the latter's toxicity can be reduced by special processing (41,42). It is difficult to conclude the safety of ingredients in each formulation individually, as numerous biological activities are likely occurring between constituent phytochemicals. Such interaction would likely see synergism or neutralisation of psoriatic activities.

CHM's effects on TNF-a reduction, suggests it may be beneficial for sufferers for which conventional pharmacological TNF-a inhibitors are unsuitable (43).

Further research is needed to investigate safety of combining oral CHM and oral conventional therapies. Health practitioners should prescribe combined therapy with caution and to reduce likelihood of unwanted effects, consider sufficient time between the administering of each.

Methodological quality of included studies

Two (22,27) of the four studies for PASI60 outcome had no blinding of participants or personnel. Such insufficiency in design may have resulted in outcome bias and contributed in part to effect size. Similarly, blinding of participants was absent for one of five studies for PASI score outcomes (22).

Only one study (22) utilised recommended treatment efficacy outcome PASI75 making it difficult to pool or compare results with other international clinical trials. No included trials included all CONSORT items (44). Such reporting would provide more detail on accuracy and specificity of each intervention for its target population. Further four studies (21,22,24,25) had incomplete data reporting in their publications, raising concern over the quality and accuracy of their reporting. It was noted final assessment data for Wang 2003 and Zhang 2009 studies had identical outcomes despite difference in initial sample size numbers and reported interventions. Age, duration of disease and gender descriptors were also identical for both studies, raising concern among reviewers of their legitimacy. No other bias or conflict of interest was identified in the studies. Future trials need to improve quality and content of their reporting. Publishing of study protocols would ensure consistency between investigated and reported outcomes.

Potential therapeutic actions of Chinese herbal medicine for psoriasis vulgaris

Despite formulation variation between studies, there were common CHM ingredients. The CM actions and biological activity of their constituents may assist explain their repeated selection in clinical trials of psoriasis.

Salvia miltiorrhiza

The CM syndrome blood stagnation is commonly attributed to psoriasis and it is likely *salvia miltiorrhiza's* (21,22,25–27) ($N = 5$) CM functions of moving blood and breaking up blood stasis is why it was most commonly utilised amongst the reviewed studies (40).

Biologically *salvia miltiorrhiza* constituents have been evidenced to suppress inflammatory mediators such as IFN-g and IL-12. Further, it may also reduce mast cell degranulation (45) and inhibit the aggregation of platelets. Proliferation of keratinocytes is a key feature of psoriatic skin. Tanshinone IIA a *salvia miltiorrhiza* constituent has been shown to inhibit keratinocyte proliferation through apoptosis (46,47).

Angelica sinensis

Having CM functions of tonifying blood, moving blood and draining wind dampness, *angelica sinensis* (23,25–27) ($n = 4$) may reduce

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psoriasis by reducing the dryness of skin (40). Ferulic acid is one of its major constituents and it contains constituents evidenced to prolong prothrombin time and inhibit platelet aggregation (48). Further, it has both antimicrobial and hepatoprotective effects. Other biological actions that may have some effect on psoriasis include its antioxidative actions and potential as a weak analgesic. There is also evidence that it can reduce inflammation and inhibit cell proliferation of tumours (45).

Rehmannia glutinosa

In *Sheng di* (raw) form *rehmannia glutinosas'* (25,26) ($n = 2$), its CM functions are generally to cool heat, cool blood to stop bleeding and tonify yin, all functions that may reduce symptoms such as Auspitz sign (pinpoint bleeding from angiogenesis) (40). When further processed, by mixing with wine and steaming, it is known as *Shu di* (22,26,27) ($n = 3$), its CM functions as a result become less cooling and its tonifying actions are enhanced. *Rehmannia glutinosa* has been evidenced to be anti-inflammatory in action, where it has been shown to inhibit histamine release and the production of TNF-a. Both of which are known to be significant contributors to inflammation. There is further evidence that it may reduce the production of the cytokine IL-1; however, its specificity to psoriasis is unclear (49). Catapol a compound found in *rehmannia glutinosa* has anti-inflammatory action, where it reduces production of pro-inflammatory mediators such as inducible nitric oxide synthase (iNOS) and TNF-a (48,50).

More experimental research is needed to evaluate which CHM are most potent as antipsoriatic therapies and to assess their toxicity. Those constituents with anti-inflammatory and/or antiproliferative effects hold most promise for psoriasis treatment, as current conventional therapies target these pathways.

Limitations

Due to variation in the primary outcome for efficacy in the studies, only the outcomes PASI60 or above and general PASI change had sufficient number of studies to be pooled and evaluated using meta-analyses. Remaining studies were evaluated separately. Any conclusion needs to consider relatively low sample sizes of studies and identified risk of bias. Three studies (21,23,27) did not detail the full ingredients of their interventions so some CHM are missing from this review.

Clinical implication of review findings

The variety of formulations and their diverse ingredients makes recommendation on which CHM are of most benefit for psoriasis difficult. Review does indicate that various forms of administration (decoction, capsule or pill) of CHM may still benefit psoriasis however the degree of the effect is unclear.

Formulations of CHM in this review appear safe for use in people with psoriasis.

Future research implications of review findings

The sample size for included studies was relatively low and likely insufficient in power to detect statistical significance for all outcomes. Future studies should utilise larger samples sizes, minimise design bias and publish study details that follow the CONSORT and its extension for herbal interventions (44,51). Formulations of this review should be explored further to evaluate comparative efficacy and their constituents' investigated for potential antipsoriatic activity. *In vitro* and *in vivo* investigation may provide further data on the effects of oral CHM and provide recommendation on safe dosage optimisation. While studies reported psoriasis symptoms according to international treatment guideline recommendations, they did not consistently report on QoL effects. Future clinical trials should include reliable QoL instruments, validated for psoriatic populations.

Conclusion

When compared with placebo control, CHM's overall effect was not significant for PASI60 or above, however for overall PASI score change, CHM was superior to placebo. The CHM form of administration appears to influence the degree of the effect; however, it is unclear which form (decoction or capsule/pill) is superior from the review. Risk of adverse events from CHM in this review is mild, however long term safety is unknown. Review was limited by lack of consistent intervention and outcome measures, as well as low participant numbers and low quality of some included studies.

More high-quality studies are needed which utilise targeted interventions, compare CHM administration forms, have psoriasis-specific outcome measures, utilise validated QoL instruments and report long-term follow-up periods.

Disclosure statement

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ORCID detail

Shefton Parker http://orcid.org/0000-0002-3132-0301

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